Construcción de un péndulo invertido con rueda de reacción inercial y diseño e implementación de un control por retroalimentación de estado*

*Note: Sub-titles are not captured in Xplore and should not be used

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***Abstract*—This document is a model and instructions for LaTeX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. *CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

***Index Terms*—component, formatting, style, styling, insert**

I. Introducción

El péndulo invertido con rueda inercial (o *Reaction Wheel Inverted Pendulum* o *RWIP* en inglés) es un sistema que permite probar las diferentes técnicas de control moderno o tradicionales. El sistema ha sido modelado por varios autores ([1], [2], [3], [4]) por diferentes métodos como las leyes Newton o de conservación de energía.

El RWIP se utiliza en varias áreas pero destaca en la robótica, especialmente en las extremidades. Como ejemplo, es la aplicación en *The Cubli: A Cube that can Jump Up and Balance* [6], que permite que un cubo mediante un salto hacia arriba permita obtener una posicion estable.

En este caso, se modeló en base a leyes de Newton y su representación en variables de estados como el diseño del control discreto e implementación, además de la construcción física del prototipo.

II. Descripción y modelamiento del RWIP

A. Descripción del sistema

El RWIP consiste en una rueda de reacción conectado a un eje, que le permite moverse libremente, mas un motor conectado al extremo con una carga acoplada a su eje, lo anterior se ilustra en la figura 1

B. Modelamiento del sistema

Para el modelamiento del sistema presentado en la figura 1, se requieren realizar algunas variaciones debido a su estructura ya implementado, como se muestra en la figura 2

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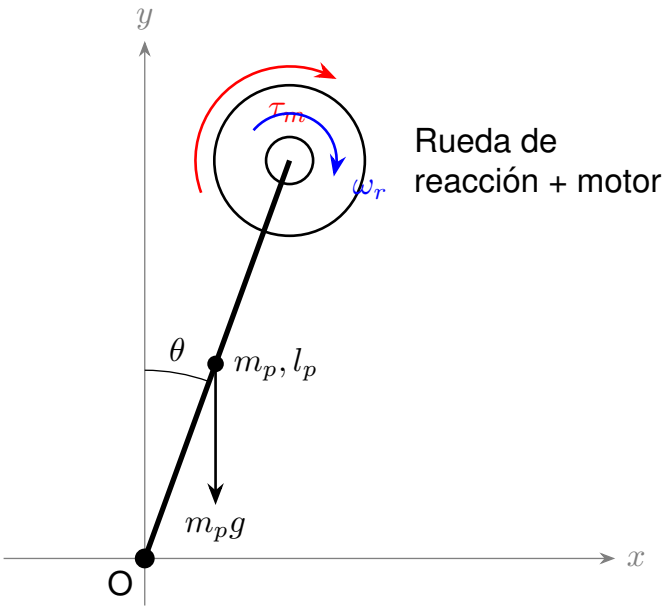


Fig. 1. Esquema del péndulo invertido con rueda de reacción (RWIP) de referencia de la guía.

1) Abreviaciones: En la figura 2 cada una de las abreviaciones corresponde a lo siguiente:

- M_m: Masa del motor.
- M_R: Masa de la rueda de reacción.
- M_b: Masa del brazo del péndulo.
- b_b: Fricción viscosa en el eje del brazo del péndulo.
- b_m: Fricción viscosa del motor.
- l_b: Distancia del centro de masa del brazo del péndulo al eje de pivote.
- l_m: Distancia del centro de masa del motor al eje de pivote.
- l_R: Distancia del centro de masa de la rueda de reacción al eje de pivote.

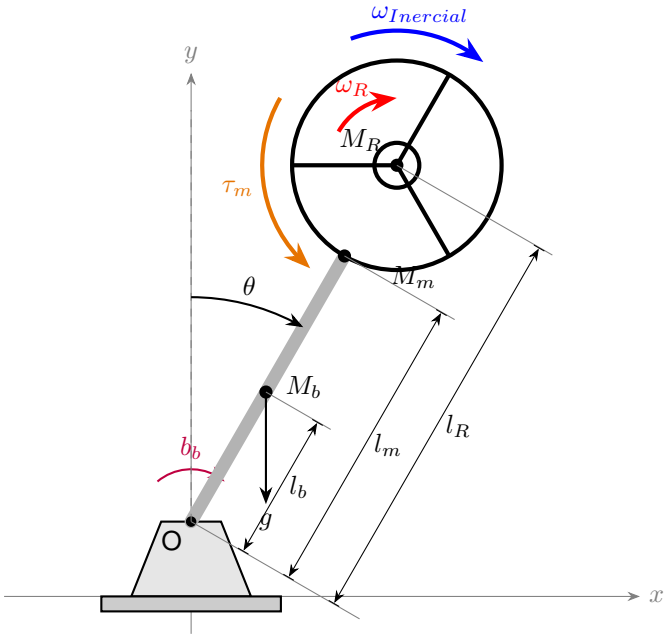


Fig. 2. Diagrama cinemático y dinámico del péndulo invertido implementado.

- g: Aceleración de la gravedad.
- τ_m: Torque de control efectivo del motor.
- I_{bo}: Inercia del péndulo con respecto al punto de origen O.
- I_{ro}: Inercia de la rueda de reacción con respecto al punto de origen O.
- I_{mo}: Inercia del motor con respecto al punto de origen O.
- I_R: Inercia de la rueda de reacción con respecto a su propio centro de masa.
- ω_R: Velocidad angular de la rueda con respecto al brazo del péndulo.
- $\omega_{Inercial}$: Velocidad angular inercial (absoluta) de la rueda.
- θ: Ángulo de inclinación del péndulo medido desde el eje vertical de referencia.
- O: Punto de pivote en la base y origen del sistema de coordenadas cartesianas.

Teniendo en cuenta la figura 2, y de acuerdo con la ecuación (1), donde I es el momento inercial y α es la aceleración angular.

$$\sum \tau = I\alpha \tag{1}$$

Se plantean las ecuaciones (3) y (2). Que representan la sumatoria del torque del motor y del brazo respectivamente.

$$\tau_b = I_t\ddot{\theta} = k\sin(\theta) - b_b\dot{\theta} - \tau_m \tag{2}$$

$$\tau_m = I_R \cdot \dot{w}_{Inercial} \tag{3}$$

Donde $w_{Incercial}$ se define en la ecuación (4) e I_R en la ecuación (5), donde r es el radio de la rueda de reacción

$$w_{Incercial} = \dot{\theta} + w_r \tag{4}$$

$$I_R = \frac{M_R r^2}{2} \tag{5}$$

Donde b_b es el coeficiente de fricción que posee la rueda y k es la magnitud del torque generado por la gravedad, se define con la ecuación (6) e I_t el momento inercial total del sistema definido en la ecuación (7)

$$k = (l_b^2 \cdot M_b + M_m \cdot l_m^2 + M_R l_R^2) \cdot g \tag{6}$$

$$I_t = I_{mo} + I_{ro} + I_{bo} \tag{7}$$

La ecuación (3), se puede escribir teniendo en cuenta el torque eléctrico (τ_e), como en la ecuación (8)

$$\tau_m = \tau_e - b_m w_r \tag{8}$$

Donde el τ_e se define en la ecuación (9), donde $i(t)$ es la corriente del motor en el tiempo y k_i es la constante de torque del motor

$$\tau_e = k_i \cdot i(t) \tag{9}$$

Teniendo todas las ecuaciones de la parte mecánica del sistema, faltan obtener las ecuaciones de la parte eléctrica que posee el motor donde su equivalente es la figura 3. Aplicando LK se obtiene la siguiente ecuación

$$v(t) = \mathrm{L}\frac{di(t)}{dt} + \mathrm{R} \cdot i(t) + e(t)$$

$$v(t) = \mathrm{L}\frac{di(t)}{dt} + \mathrm{R} \cdot i(t) + k_e w_r \tag{10}$$

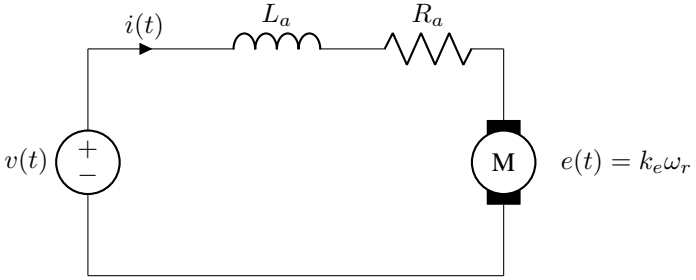


Fig. 3. Circuito eléctrico equivalente del motor de corriente continua.

C. Linealización del modelo

Como la ecuación (6) es un ecuación no lineal, se linealiza para valores θ muy pequeños, esto permite tener un rango de valores para cual se cumple la siguiente relación

$$\sin\theta \approx \theta \tag{11}$$

Esto se puede comprobar en las gráficas 4 y 5, que tiene valores lineales de la ecuación (11) comparada con la $\sin\theta$ real, donde el intervalo $[-31.6, 31.6]°$ o $[-0.55, 0.55]$ rad, son valores que tienen errores menores al 5%, lo cual hace mas exacto la linealización

Teniendo en cuenta la linealidad de la ecuación (11), se puede reescribir la ecuacion (6) como

$$I_t\ddot{\theta} = k\theta - b_b\dot{\theta} - \tau_m \tag{12}$$

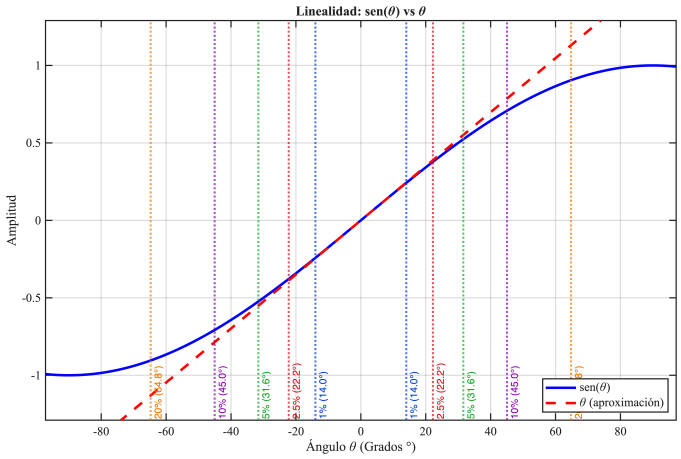


Fig. 4. Valores para cual $\sin\theta$ es lineal grados

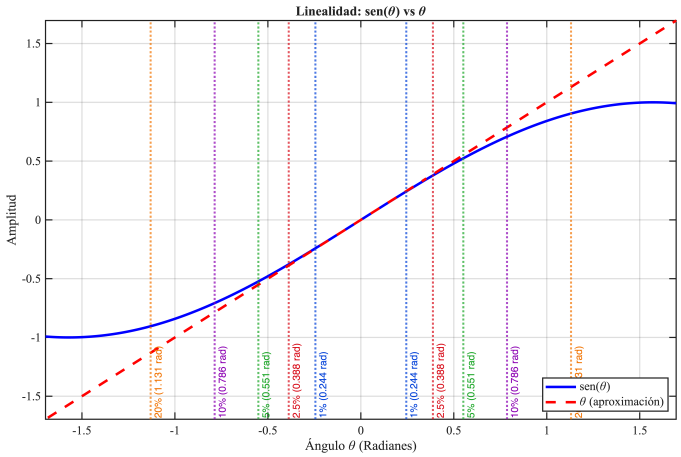


Fig. 5. Valores para cual $\sin\theta$ es lineal radianes

D. Variables de estado del modelo

De las ecuaciones obtenidas en la sección de modelamiento y de la linealización del sistema, se pueden obtener las siguientes variables de estados, que corresponden a los elementos que almacenan energía

1) w_r. Energía cinética de la rueda de reacción, velocidad angular de la rueda de reacción
2) θ. Energía cinética del sistema, posición angular del sistema.
3) $\dot{\theta}$. Energía potencial debido a la acción gravitacional. Velocidad angular del sistema.
4) I. Energía en forma de campo magnético que almacena la bobina. Corriente del motor.

Teniendo lo anterior, se obtuvo las siguientes variables de estado

$$\underline{x} = \begin{bmatrix} \theta & \dot{\theta} & w_r & I \end{bmatrix}^T \tag{13}$$

Donde la entrada es:

$$u(t) = v(t) \tag{14}$$

1) Obtención de $\dot{x}_1$: Se puede obtener la relación de la ecuación (15), debido a que $\dot{\theta}$ es la dervida de θ

$$\dot{x}_1 = x_2 \tag{15}$$

2) Obtención de $\dot{x}_2$: De la ecuación (12), se reescribe con variables de estado y despeja $\dot{x}_2$ dando como resultado la ecuación (16)

$$\dot{x}_2 = \frac{k}{I_t}x_1 - \frac{b_b}{I_t}x_2 - \frac{\tau_m}{I_t} \tag{16}$$

Se reemplaza la ecuación (9) en (8) y luego esta en la ecuación (16). Todo lo anterior con variable de estados.

$$\tau_e = k_i x_4$$

$$\tau_m = k_i x_4 - b_m x_3 \tag{17}$$

$$\dot{x}_2 = \frac{k}{I_t}x_1 - \frac{b_b}{I_t}x_2 - \frac{k_i x_4 - b_m x_3}{I_t}$$

$$\dot{x}_2 = \frac{k}{I_t}x_1 - \frac{b_b}{I_t}x_2 + \frac{b_m}{I_t}x_3 - \frac{k_i}{I_t}x_4 \tag{18}$$

3) Obtención de $\dot{x}_3$: Se reemplaza las ecuaciones (17) y (4) en la ecuación (3), esta ultima se reemplaza en la ecuación (18) y por último se despeja $\dot{x}_3$ dando como resultado la ecuación (19)

$$k_i x_4 - b_m x_3 = I_R \cdot (\dot{x}_2 + \dot{x}_3)$$

$$\dot{x}_3 = -\dot{x}_2 - \frac{b_m}{I_R}x_3 + \frac{k_i}{I_R}x_4$$

$$\dot{x}_3 = -\left(\frac{k}{I_t}x_1 - \frac{b_b}{I_t}x_2 + \frac{b_m}{I_t}x_3 - \frac{k_i}{I_t}x_4\right) - \frac{b_m}{I_R}x_3 + \frac{k_i}{I_R}x_4$$

$$\dot{x}_3 = -\frac{k}{I_t}x_1 + \frac{b_b}{I_t}x_2 - b_m\frac{I_R + I_t}{I_R I_t}x_3 + \frac{k_i}{I_R}x_4 \tag{19}$$

4) Obtención de $\dot{x}_4$: Se reescribe la ecuacion (10) en variables de estado y se despeja $\dot{x}_4$

$$u(t) = \mathrm{L}_a\dot{x}_4 + \mathrm{R}x_4 + k_e x_3$$

$$\dot{x_4} = \frac{k_e}{\mathrm{L}_a}x_3 + \frac{\mathrm{R}_a}{\mathrm{L}_a}x_4 - \frac{1}{\mathrm{L}_a}u(t) \tag{20}$$

Teniendo en cuenta cada una de las variables de estados, se arma la matriz de estado, de acuerdo a la ecuación (21)

$$\begin{aligned} \dot{\underline{x}}(t) &= A\underline{x}(t) + Bu(t) \\ \underline{y}(t) &= C\underline{x}(t) + Du(t) \end{aligned} \tag{21}$$

Donde las matrices A, B, C y D corresponden a las ecuaciones (22), (23), (24) y (25) respectivamente.

$$A = \begin{bmatrix} 0 & 1 & 0 & 0 \\ \frac{k}{I_t} & -\frac{b_b}{I_t} & \frac{b_m}{I_t} & -\frac{k_i}{I_t} \\ -\frac{k}{I_t} & \frac{b_b}{I_t} & -b_m\frac{I_R+I_t}{I_R I_t} & \frac{k_i}{I_R} \\ 0 & 0 & \frac{k_e}{\mathrm{L}_a} & \frac{\mathrm{R}_a}{\mathrm{L}_a} \end{bmatrix} \tag{22}$$

$$B = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -\frac{1}{\mathrm{L}_a} \end{bmatrix} \tag{23}$$

$$C = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{24}$$

$$D = 0 \tag{25}$$

III. Prepare Your Paper Before Styling

Before you begin to format your paper, first write and save the content as a separate text file. Complete all content and organizational editing before formatting. Please note sections III-A–III-E below for more information on proofreading, spelling and grammar.

Keep your text and graphic files separate until after the text has been formatted and styled. Do not number text heads—LaTeX will do that for you.

A. Abbreviations and Acronyms

Define abbreviations and acronyms the first time they are used in the text, even after they have been defined in the abstract. Abbreviations such as IEEE, SI, MKS, CGS, ac, dc, and rms do not have to be defined. Do not use abbreviations in the title or heads unless they are unavoidable.

B. Units

- Use either SI (MKS) or CGS as primary units. (SI units are encouraged.) English units may be used as secondary units (in parentheses). An exception would be the use of English units as identifiers in trade, such as "3.5-inch disk drive".
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- Use a zero before decimal points: "0.25", not ".25". Use "cm^3", not "cc".)

C. Equations

Number equations consecutively. To make your equations more compact, you may use the solidus (/), the exp function, or appropriate exponents. Italicize Roman symbols for quantities and variables, but not Greek symbols. Use a long dash rather than a hyphen for a minus sign. Punctuate equations with commas or periods when they are part of a sentence, as in:

$$a + b = \gamma \tag{26}$$

Be sure that the symbols in your equation have been defined before or immediately following the equation. Use "(26)", not "Eq. (26)" or "equation (26)", except at the beginning of a sentence: "Equation (26) is . . ."

D. LaTeX-Specific Advice

Please use "soft" (e.g., `\eqref{Eq}`) cross references instead of "hard" references (e.g., `(1)`). That will make it possible to combine sections, add equations, or change the order of figures or citations without having to go through the file line by line.

Please don't use the `{eqnarray}` equation environment. Use `{align}` or `{IEEEeqnarray}` instead. The `{eqnarray}` environment leaves unsightly spaces around relation symbols.

Please note that the `{subequations}` environment in LaTeX will increment the main equation counter even when there are no equation numbers displayed. If you forget that, you might write an article in which the equation numbers skip from (17) to (20), causing the copy editors to wonder if you've discovered a new method of counting.

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Do not use `\nonumber` inside the `{array}` environment. It will not stop equation numbers inside `{array}` (there won't be any anyway) and it might stop a wanted equation number in the surrounding equation.

E. Some Common Mistakes

- The word "data" is plural, not singular.
- The subscript for the permeability of vacuum μ_0, and other common scientific constants, is zero with subscript formatting, not a lowercase letter "o".
- In American English, commas, semicolons, periods, question and exclamation marks are located within quotation marks only when a complete thought or name is cited,

such as a title or full quotation. When quotation marks are used, instead of a bold or italic typeface, to highlight a word or phrase, punctuation should appear outside of the quotation marks. A parenthetical phrase or statement at the end of a sentence is punctuated outside of the closing parenthesis (like this). (A parenthetical sentence is punctuated within the parentheses.)

- A graph within a graph is an "inset", not an "insert". The word alternatively is preferred to the word "alternately" (unless you really mean something that alternates).
- Do not use the word "essentially" to mean "approximately" or "effectively".
- In your paper title, if the words "that uses" can accurately replace the word "using", capitalize the "u"; if not, keep using lower-cased.
- Be aware of the different meanings of the homophones "affect" and "effect", "complement" and "compliment", "discreet" and "discrete", "principal" and "principle".
- Do not confuse "imply" and "infer".
- The prefix "non" is not a word; it should be joined to the word it modifies, usually without a hyphen.
- There is no period after the "et" in the Latin abbreviation "et al.".
- The abbreviation "i.e." means "that is", and the abbreviation "e.g." means "for example".

An excellent style manual for science writers is [**?**].

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Headings, or heads, are organizational devices that guide the reader through your paper. There are two types: component heads and text heads.

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Text heads organize the topics on a relational, hierarchical basis. For example, the paper title is the primary text head because all subsequent material relates and elaborates on this one topic. If there are two or more sub-topics, the next level head (uppercase Roman numerals) should be used and, conversely, if there are not at least two sub-topics, then no subheads should be introduced.

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a) Positioning Figures and Tables: Place figures and tables at the top and bottom of columns. Avoid placing them in the middle of columns. Large figures and tables may span across both columns. Figure captions should be below the figures; table heads should appear above the tables. Insert figures and tables after they are cited in the text. Use the abbreviation "Fig. 6", even at the beginning of a sentence.

TABLE I
TABLE TYPE STYLES

Table Head	**Table Column Head**		
	Table column subhead	***Subhead***	***Subhead***
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[a]Sample of a Table footnote.



Fig. 6. Example of a figure caption.

Figure Labels: Use 8 point Times New Roman for Figure labels. Use words rather than symbols or abbreviations when writing Figure axis labels to avoid confusing the reader. As an example, write the quantity "Magnetization", or "Magnetization, M", not just "M". If including units in the label, present them within parentheses. Do not label axes only with units. In the example, write "Magnetization (A/m)" or "Magnetization {A[m(1)]}", not just "A/m". Do not label axes with a ratio of quantities and units. For example, write "Temperature (K)", not "Temperature/K".

ACKNOWLEDGMENT

The preferred spelling of the word "acknowledgment" in America is without an "e" after the "g". Avoid the stilted expression "one of us (R. B. G.) thanks ...". Instead, try "R. B. G. thanks...". Put sponsor acknowledgments in the unnumbered footnote on the first page.

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Please number citations consecutively within brackets [**?**]. The sentence punctuation follows the bracket [**?**]. Refer simply to the reference number, as in [**?**]—do not use "Ref. [**?**]" or "reference [**?**]" except at the beginning of a sentence: "Reference [**?**] was the first ..."

Number footnotes separately in superscripts. Place the actual footnote at the bottom of the column in which it was cited. Do not put footnotes in the abstract or reference list. Use letters for table footnotes.

Unless there are six authors or more give all authors' names; do not use "et al.". Papers that have not been published, even if they have been submitted for publication, should be cited as "unpublished" [**?**]. Papers that have been accepted for publication should be cited as "in press" [**?**]. Capitalize only the first word in a paper title, except for proper nouns and element symbols.

For papers published in translation journals, please give the English citation first, followed by the original foreign-language citation [**?**].

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